QUESTION ANALYSIS

# EEE337

Power System II

# Question Analysis

**Sessions 2021–22 to 2020–21** *2 papers, 51 questions, 96 formulas*

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# 1. At a glance:

This analysis covers **2 papers** (2021-22, 2020-21), with 51 questions and 325 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

### The five topics with the most marks:

| Topic | Marks | Share | Questions | Top subtopics (marks) |
|---|---|---|---|---|
| Underground cables | 85 | 26 % | 11 | Dielectric stress (43); Grading (20); Capacitance (10) |
| Line inductance | 66 | 20 % | 11 | Mutual inductance (25); Composite conductors (18); Transposition (12) |
| Overhead insulators | 48 | 15 % | 7 | String efficiency (41); Failure (7) |
| Power system stability | 43 | 13 % | 5 | Equal area criterion (23); Transient EMF (20) |
| Line capacitance | 27 | 8 % | 5 | Unsymmetrical spacing (25); Effect of earth (2) |

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

# 2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

| Topic and subtopic | 2021-22 | 2020-21 | Term tests | All |
|---|---|---|---|---|
| **Line inductance** | **26** | **30** | **10** | **66** |
| Mutual inductance | **10** | **10** | **5** | 25 |
| Composite conductors | **8** | **10** | · | 18 |
| Transposition | **8** | **4** | · | 12 |
| Bundled conductors | · | **4** | **5** | 9 |
| Basics | · | **1** | · | 1 |
| Skin effect | · | **1** | · | 1 |
| **Line capacitance** | **17** | **10** | · | **27** |
| Unsymmetrical spacing | **15** | **10** | · | 25 |
| Effect of earth | **2** | · | · | 2 |
| **Corona** | **2** | **6** | · | **8** |
| Critical disruptive voltage | **2** | **5** | · | 7 |
| Corona effect | · | **1** | · | 1 |
| **Overhead insulators** | **28** | **15** | **5** | **48** |
| String efficiency | **26** | **10** | **5** | 41 |
| Failure | **2** | **5** | · | 7 |
| **Mechanical design** | **7** | **10** | · | **17** |
| Sag | **2** | **10** | · | 12 |
| Supports | **5** | · | · | 5 |
| **Underground cables** | **40** | **40** | **5** | **85** |
| Dielectric stress | **8** | **30** | **5** | 43 |
| Grading | **20** | · | · | 20 |
| Capacitance | · | **10** | · | 10 |
| Construction | **5** | · | · | 5 |
| Insulation resistance | **5** | · | · | 5 |

| Topic and subtopic | 2021-22 | 2020-21 | Term tests | All |
|---|---|---|---|---|
| Economical size | **2** | · | · | 2 |
| **Cable faults** | **5** | **10** | **5** | **20** |
| Loop tests | **5** | **10** | **5** | 20 |
| **Power system stability** | **18** | **25** | · | **43** |
| Equal area criterion | **8** | **15** | · | 23 |
| Transient EMF | **10** | **10** | · | 20 |
| **HVDC and FACTS** | **7** | **4** | · | **11** |
| HVDC | **7** | · | · | 7 |
| FACTS | · | **4** | · | 4 |

# 3. Priority list:

**Must know:** Underground cables (85), Line inductance (66), Overhead insulators (48), Power system stability (43), Line capacitance (27), Cable faults (20), Mechanical design (17), HVDC and FACTS (11), Corona (8)

**Likely:** none

**Less often:** none

**The rule:** Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

# 4. Topic pages:

## 4.1. Line inductance:

66 marks, 11 questions, in 2 of 2 papers. Book: Mehta Ch. 9; Grainger Ch. 4.

Most often numerical (5 of 11). Subtopics: Basics; Bundled conductors; Mutual inductance; Transposition; Composite conductors; Skin effect.

**Formulas:**

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Basics (1) | **Inductance** | $L = \frac{\psi}{I} = \frac{N\phi}{I}$ | 20-21 1(a) |
| | Induced voltage | $e = L\frac{di}{dt}$ | 20-21 1(a) |
| Bundled conductors (9) | Internal flux linkage | $\psi_{int} = \frac{\mu_0 I}{8\pi}$ | 21-22 1(c), 20-21 1(b) |
| | External flux linkage | $\psi_{12} = \frac{\mu_0 I}{2\pi}\ln\frac{D_2}{D_1} = 2\times10^{-7}\, I \ln\frac{D_2}{D_1}$ | 20-21 1(b) |
| | **GMR of a solid conductor** | $D_s = r' = r\, e^{-1/4} = 0.7788\, r$ | 20-21 1(b) |
| | **Transposed 3-phase line** | $L = 2\times10^{-7}\ln\frac{D_{eq}}{D_s}$ H/m | 20-21 1(b) |
| | **Equivalent spacing** | $D_{eq} = \sqrt[3]{D_{ab}D_{bc}D_{ca}}$ | 20-21 1(b) |

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | **Two-conductor bundle** | $D_s^b = \sqrt{D_s\, d}$ | 20-21 1(b) |
| | **Inductive reactance** | $X_L = 2\pi f L$ | 21-22 7(a), 20-21 1(b) |
| | Two-conductor bundle | $D_s = \sqrt{r'\, d}$ | 21-22 7(a) |
| Mutual inductance (25) | **Mutual inductance** | $M = 2\times10^{-7}\ln\dfrac{D_{P_1T_2}\,D_{P_2T_1}}{D_{P_1T_1}\,D_{P_2T_2}}$ H/m | 21-22 1(d), 20-21 1(d) |
| | Induced voltage | $E = 2\pi f M I$ | 21-22 1(d), 20-21 1(d) |
| | Symmetrical arrangement | $M = 4\times10^{-7}\ln\dfrac{D_2}{D_1}$ H/m | 21-22 1(d) |
| Transposition (12) | Flux linkage in a group | $\psi_a = 2\times10^{-7}\left[I_a\ln\dfrac{1}{D_s} + I_b\ln\dfrac{1}{D_{ab}} + I_c\ln\dfrac{1}{D_{ac}}\right]$ | 20-21 2(b) |
| | Inductance | $L = \dfrac{\psi}{I}$ | 21-22 1(c) |
| | External flux linkage | $\psi_{12} = \dfrac{\mu_0 I}{2\pi}\ln\dfrac{D_2}{D_1}$ | 21-22 1(c) |
| | **GMR of a solid conductor** | $r' = r\,e^{-1/4} = 0.7788\,r$ | 21-22 1(c) |
| | Symmetrical 3-phase line | $L = 2\times10^{-7}\ln\dfrac{D}{r'}$ H/m | 21-22 1(c) |
| | **Unsymmetrical transposed line** | $L = 2\times10^{-7}\ln\dfrac{D_{eq}}{r'} = 0.2\ln\dfrac{D_{eq}}{r'}$ mH/km | 21-22 1(c) |
| | **Equivalent spacing** | $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$ | 21-22 1(c) |
| Composite conductors (18) | **Composite conductor** | $L_X = 2\times10^{-7}\ln\dfrac{D_m}{D_{sX}}$ H/m | 21-22 4(c), 20-21 2(c) |
| | Mutual GMD | $D_m = \sqrt[mn]{(D_{aa'}D_{ab'}\cdots D_{am})\cdots(D_{na'}\cdots D_{nm})}$ | 21-22 4(c), 20-21 2(c) |
| | Self GMD | $D_{sX} = \sqrt[n^2]{(D_{aa}D_{ab}\cdots D_{an})\cdots(D_{na}\cdots D_{nn})}$ | 21-22 4(c), 20-21 2(c) |
| | Strand inductance | $L_X = \dfrac{L_a + L_b + \cdots + L_n}{n^2}$ | 20-21 2(c) |
| | Composite line loop | $L = 2\times10^{-7}\ln\dfrac{D_m^2}{D_{sX}D_{sY}}$ H/m | 21-22 4(c), 20-21 2(c) |
| Skin effect (1) | Skin depth | $\delta = \dfrac{1}{\sqrt{\pi f\mu\sigma}}$ | 20-21 3(a) |


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## Symbols:

$\psi$ = flux linkage (Wb-T/m) ↩

$\phi$ = flux (Wb) ↩

$I$ = current (A) ↩

$e$ = self-induced voltage (V) ↩

$D_1$, $D_2$ = distances of two points from the conductor (m); near and far distances (m) ↩

$d$ = spacing of the sub-conductors (m) ↩

$f$ = frequency (Hz) ↩

$D_{P_iT_j}$ = distance from power wire $i$ to telephone wire $j$ (m)

$E$ = voltage in the telephone line (V per m) ↩

$r'$ = geometric mean radius (m) ↩

$D$ = equal spacing of the phases (m) ↩

$D_{12}$, $D_{23}$, $D_{31}$ = the three phase spacings (m) ↩

$D_m$ = mutual GMD between conductors X and Y (m) ↩

$D_{sX}$ = self GMD of X (m) ↩

$n$, $m$ = strands in X and Y ↩

$\sigma$ = conductivity (S/m) ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Basics** | | | | |
| 1 | 20-21 1(a) | 1 | theory | Define Inductance with the required expression. |
| **Bundled conductors** | | | | |
| 2 | 21-22 TT1 7(a) | 5 | numerical | A 500 kV line has a bundling arrangement of 2 conductors per phase as shown in the Figure 1. Compute the inductive reactance per phase of this line at 50 Hz. Each conductor carries 50% of the phase current. |
| 3 | 20-21 1(b) | 4 | numerical | Each conductor of the bundled-conductor line shown in Fig.1 is ACSR, 1,272,000-cmil Pheasant. Find the inductive reactance in ohms per km (and per mile) per phase for d=45cm. |
| **Mutual inductance** | | | | |
| 4 | 21-22 1(d) | 10 | numerical | A single-phase 50 Hz power line is supported on a horizontal cross-arm. The spacing between the conductors is 3 m. A telephone line is supported symmetrically below the power line as shown in Fig. 1. Find the mutual inductance between the two circuits and the voltage induced per kilometer in the telephone line if the current in the power line is 100 A. Assume the telephone line current to be zero. |
| 5 | 21-22 TT3 9(b) | 5 | numerical repeats 20-21 1(d) | A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. |
| 6 | 20-21 1(d) | 10 | numerical | A single phase 50Hz power line is supported on a horizontal cross-arm. The spacing between conductors is 2.5m. A telephone line is also supported on a horizontal cross-arm in the same horizontal plane as the power line. The conductors of the telephone line are of solid copper spaced 0.6m between centers. The distance between the nearest conductors of the two lines is 20m. Find the mutual inductance between the circuits and the voltage per km induced in the telephone line for 150A current flowing over the power line. |
| **Transposition** | | | | |
| 7 | 21-22 1(c) | 8 | derivation | Define transpose transmission line. Derive the expression for the inductance of three phase transmission lines with unsymmetrical spacing. |
| 8 | 20-21 2(b) | 4 | theory | Define Transposed Transmission line. Explain the reason behind Transpose. |
| **Composite conductors** | | | | |
| 9 | 21-22 4(c) | 8 | derivation repeats 20-21 2(c) | Derive the expression of inductance of composite conductor lines. |
| 10 | 20-21 2(c) | 10 | derivation | Derive the expression of Inductance of composite conductor lines. |
| **Skin effect** | | | | |
| 11 | 20-21 3(a) | 1 | theory | Define Skin Effect. |


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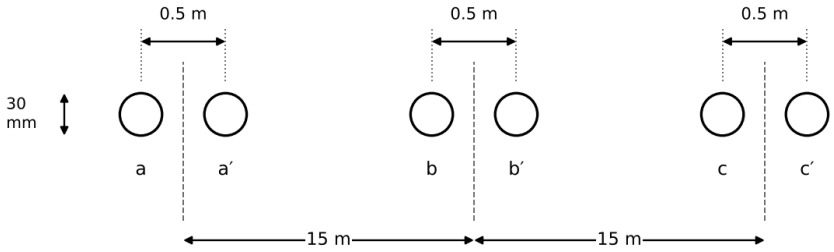



**Figure 1:** 21-22 TT1 7(a): Figure 1 of Term Test 1 (redrawn)

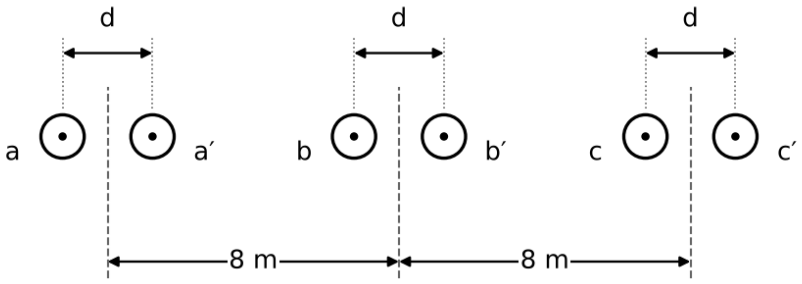



**Figure 2:** 20-21 1(b): Fig. 1 of the paper (redrawn): figure for Q-1(b)

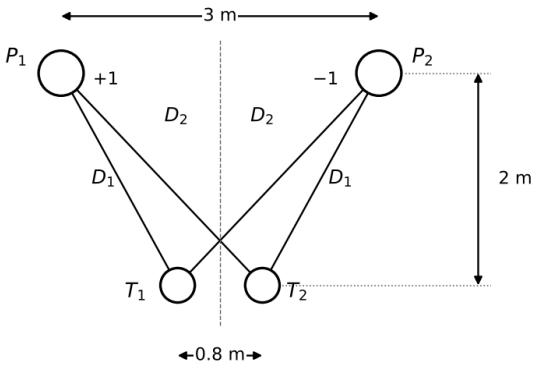



**Figure 3:** 21-22 1(d): Fig. 1 of the paper (redrawn): figure for Q-1(d)


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## 4.2. Line capacitance:

27 marks, 5 questions, in 2 of 2 papers. Book: Mehta Ch. 9; Grainger Ch. 5.

Most often derivation (3 of 5). Subtopics: Unsymmetrical spacing; Effect of earth.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Unsymmetrical spacing (25) | Capacitance | $C = \frac{q}{V}$ | 20-21 1(c) |
| | Field of a line charge | $E = \frac{q}{2\pi\varepsilon_0\, x}$ | 20-21 1(c) |
| | Potential difference | $V_{12} = \frac{q}{2\pi\varepsilon_0} \ln \frac{D_2}{D_1}$ | 21-22 5(d), 20-21 1(c) |
| | Balanced voltages | $V_{ab} + V_{ac} = 3V_{an}$ | 20-21 1(c) |
| | **Capacitance to neutral** | $C_n = \frac{2\pi\varepsilon_0}{\ln(D_{eq}/r)} = \frac{0.0556}{\ln(D_{eq}/r)}\ \mu\text{F/km}$ | 20-21 1(c) |
| | Line charging current | $I_c = 2\pi f\, C_n\, V_{ph}$ | 21-22 5(b), 20-21 1(c) |
| | Field of a line charge | $g = \frac{q}{2\pi\varepsilon_0\, x}$ | 21-22 5(d) |
| | **Capacitance to neutral** | $C_n = \frac{2\pi\varepsilon_0}{\ln(D_{eq}/r)}\ \text{F/m}$ | 21-22 5(d) |
| Effect of earth (2) | Single-phase line | $C = \frac{\pi\varepsilon_0}{\ln(d/r)}\ \text{F/m}$ | 21-22 5(a) |
| | Effect of the earth | $C = \frac{\pi\varepsilon_0}{\ln\left(\frac{d}{r} \cdot \frac{2h}{\sqrt{4h^2 + d^2}}\right)}$ | 21-22 5(a) |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|

### Symbols:

$q$ = charge (C/m) ↩ $V_{an}$ = phase voltage ↩ $d$ = spacing of the two wires (m) ↩

$x$ = distance from the conductor (m) ↩ $V_{ph}$ = phase voltage (V) ↩ $h$ = height above the earth (m) ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Unsymmetrical spacing** | | | | |
| 1 | 21-22 5(b) | 5 | numerical | A three phase 50 Hz transmission line has flat horizontal spacing with 3.5 m between adjacent conductors. The conductors are No. 2/0 hard/drawn seven strand copper (outside conductor diameter= 1.05 cm). The voltage of the line is 110 kV. Find the capacitance to neutral and the charging current per kilometre of line. |
| 2 | 21-22 5(d) | 10 | derivation repeats 20-21 1(c) | Derive the expression of capacitance of a three phase line with unsymmetrical spacing. |
| 3 | 21-22 TT1 7(b) | 0 | derivation repeats 21-22 5(d) | Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing. |
| 4 | 20-21 1(c) | 10 | derivation | Derive the expression of Capacitance of a three phase line with unsymmetrical spacing. |
| **Effect of earth** | | | | |
| 5 | 21-22 5(a) | 2 | theory | How Earth affect on transmission line capacitance? |

## 4.3. Corona:

8 marks, 3 questions, in 2 of 2 papers. Book: Mehta Ch. 8.

Most often theory (3 of 3). Subtopics: Corona effect; Critical disruptive voltage.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Corona effect (1) | Field near the conductor | $g = \dfrac{V}{x \ln(d/r)}$ | 21-22 1(a), 20-21 2(a) |
| | Strength of air | $g_0 = 30$ kV/cm peak $= 21.2$ kV/cm r.m.s. | 21-22 1(a), 20-21 2(a) |
| Critical disruptive voltage (7) | Surface field | $g_{max} = \dfrac{V}{r \ln(d/r)}$ | 21-22 1(a), 20-21 4(a) |
| | **Critical disruptive voltage** | $V_c = m_0\, g_0\, \delta\, r \ln \dfrac{d}{r}$ kV | 21-22 1(a), 20-21 4(a) |
| | Air density factor | $\delta = \dfrac{3.92\, b}{273 + t}$ | 21-22 1(a), 20-21 4(a) |
| | Visual critical voltage | $V_v = m_v\, g_0\, \delta\, r \left(1 + \dfrac{0.3}{\sqrt{\delta r}}\right) \ln \dfrac{d}{r}$ | 20-21 4(a) |



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### Symbols:

$V$ = phase-to-neutral voltage (kV) ↩

$d$ = spacing (cm) ↩

$g_0$ = breakdown strength of air ↩

$m_0$ = irregularity factor (1 smooth, 0.80–0.87 stranded) ↩

$r$ = conductor radius (cm) ↩

$b$ = pressure (cm of Hg) ↩

$t$ = temperature (°C) ↩

$m_v$ = irregularity factor for visual corona ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Corona effect** | | | | |
| 1 | 20-21 2(a) | 1 | theory | Define Corona Effect. |
| **Critical disruptive voltage** | | | | |
| 2 | 21-22 1(a) | 2 | theory repeats 20-21 4(a) | Define critical disruptive voltage. Write down it's significance. |
| 3 | 20-21 4(a) | 5 | theory | Define critical disruptive voltage. Write down its significance. |

## 4.4. Overhead insulators:

48 marks, 7 questions, in 2 of 2 papers. Book: Mehta Ch. 8.

Most often theory (3 of 7). Subtopics: String efficiency; Failure.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| String efficiency (41) | **Capacitance ratio** | $K = \dfrac{C_1}{C}$ | 21-22 3(c), 20-21 4(b) |
| | **Unit voltages** | $V_2 = (1+K)V_1,\ V_3 = (1+3K+K^2)V_1$ | 20-21 4(b) |
| | General unit | $V_{k+1} = V_k + K\,(V_1 + V_2 + \cdots + V_k)$ | 21-22 8(a), 20-21 4(b) |
| | **String efficiency** | $\eta = \dfrac{V}{n\,V_n} \times 100\ \%$ | 21-22 3(d), 20-21 4(b) |
| | Three-unit string | $\eta = \dfrac{3+4K+K^2}{3(1+3K+K^2)}$ | 21-22 3(d), 20-21 4(b) |
| | **Capacitance grading** | $C_2 = C_1(1+K),\ C_3 = C_1(1+3K)$ | 20-21 4(b) |
| | **Unit voltages** | $V_2 = V_1(1+K),\ V_3 = V_2 + K(V_1+V_2)$ | 21-22 3(c) |
| | String voltage | $V = V_1 + V_2 + \cdots + V_n$ | 21-22 3(c) |
| | $K$ from measurements | $K = \dfrac{V_2}{V_1} - 1$ | 21-22 5(c) |
| Failure (7) | Safety factor of an insulator | $\text{SF} = \dfrac{\text{puncture strength}}{\text{flash-over voltage}}$ | 21-22 3(a), 20-21 5(a) |

### Symbols:

$C_1$ = shunt capacitance to the tower (F); shunt capacitance to earth (F) ↩

$C$ = self capacitance of a unit (F) ↩

$V_1$ = voltage of the top (cross-arm) unit (V) ↩

$V$ = phase voltage across the string (V) ↩

$n$ = number of units ↩

$V_n$ = voltage of the unit nearest the line (V) ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **String efficiency** | | | | |
| 1 | 21-22 3(c) | 8 | numerical | In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self capacitance of each insulator, find (i) the distribution of voltage over 3 insulators and (ii) String efficiency. |
| 2 | 21-22 3(d) | 10 | derivation | Derive the mathematical expression of String efficiency with proper diagram. What will happen if: (i) $K = 0.2$ ($K = \frac{shunt\ capacitance}{self\ capacitance}$) (ii) $K = 0.8$ (iii) the number of disc increases in the string. |
| 3 | 21-22 5(c) | 8 | numerical | A 3-phase transmission line is being supported by three-disc insulators. The potentials across top unit (near to the tower) and middle unit are 8 kV and 11 kV, respectively. (i) Draw the equivalent circuit (ii) Calculate the ratio of capacitance between pin and earth to the self-capacitance of each unit (iii) Calculate the line voltage and (iv) Calculate string efficiency. |
| 4 | 21-22 TT2 8(a) | 5 | numerical | A string of 5 insulators is connected across a 100 kV line. If the capacitance of each disc to earth is 0.1 of the capacitance of the insulator, calculate (i) the distribution of voltage on the insulator discs and (ii) the string efficiency. |
| 5 | 20-21 4(b) | 10 | theory | What is String Efficiency? Discuss the methods of improving String Efficiency. |
| **Failure** | | | | |
| 6 | 21-22 3(a) | 2 | theory | How electrical breakdown happen in a pin-type insulator? |
| 7 | 20-21 5(a) | 5 | theory | What are the causes of insulation failure? |

## 4.5. Mechanical design:

17 marks, 3 questions, in 2 of 2 papers. Book: Mehta Ch. 8.

Most often theory (2 of 3). Subtopics: Sag; Supports.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Sag (12) | **Working tension** | $T = \dfrac{\text{ultimate strength}}{\text{safety factor}}$ | 20-21 5(c) |
| | **Sag, level supports** | $S = \dfrac{w\,l^2}{8\,T}$ | 21-22 4(a), 20-21 5(c) |
| | Conductor curve | $y = \dfrac{w\,x^2}{2\,T}$ | 20-21 5(c) |
| | **Height of supports** | $h = \text{ground clearance} + S$ | 20-21 5(c) |

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|

### Symbols:

$w$ = weight per metre (N/m) ↩ $T$ = tension (same force unit as $w$) ↩ $x$ = distance from the lowest point (m) ↩

$l$ = span (m) ↩ $S$ = sag (m) ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Sag** | | | | |
| 1 | 21-22 4(a) | 2 | theory | What is Stringing charts? |
| 2 | 20-21 5(c) | 10 | numerical | A 132 KV transmission line has the following data: Weight of the conductor= 680 kg/km, Length of span= 260 m, Ultimate strength= 3100 kg, Safety Factor= 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 meters. |
| **Supports** | | | | |
| 3 | 21-22 3(b) | 5 | theory | What kind of line supports are used for distribution purposes at low voltages (upto 11 kV)? |

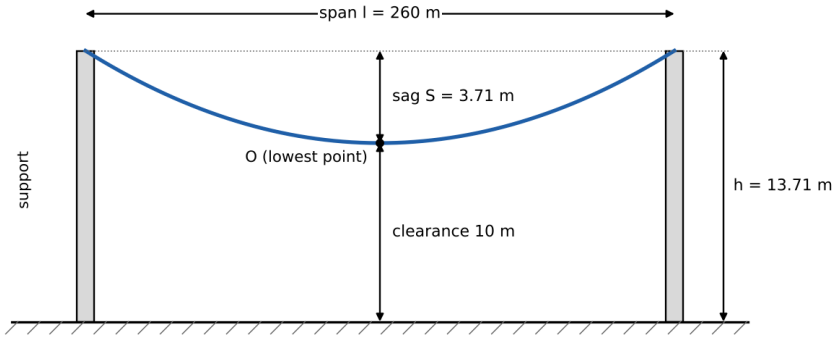



**Figure 4:** 20-21 5(c): supports at the same level, sag S and clearance

## 4.6. Underground cables:

85 marks, 11 questions, in 2 of 2 papers. Book: Mehta Ch. 11.

Most often numerical (8 of 11). Subtopics: Dielectric stress; Capacitance; Construction; Grading; Insulation resistance; Economical size.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Dielectric stress (43) | Flux density in the dielectric | $D_x = \dfrac{Q}{2\pi x}$ | 20-21 3(c) |
| | Stress from the charge | $g_x = \dfrac{Q}{2\pi\varepsilon_0\varepsilon_r\, x}$ | 20-21 3(c) |
| | Core-to-sheath voltage | $V = \dfrac{Q}{2\pi\varepsilon_0\varepsilon_r} \ln \dfrac{D}{d}$ | 20-21 3(c) |
| | **Dielectric stress** | $g_x = \dfrac{V}{x\, \ln(D/d)}$ | 20-21 3(c) |
| | **Maximum and minimum stress** | $g_{max} = \dfrac{2V}{d\, \ln(D/d)},\ g_{min} = \dfrac{2V}{D\, \ln(D/d)}$ | 21-22 2(c), 20-21 2(d) |

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| | **Stress ratio** | $\frac{g_{max}}{g_{min}} = \frac{D}{d}$ | 20-21 4(c) |
| | Insulation thickness | $t = \frac{D-d}{2}$ | 20-21 4(c) |
| | Operating voltage | $V = g_{max}\frac{d}{2}\ln\frac{D}{d}$ | 20-21 4(c) |
| | Cable capacitance | $C = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(D/d)}$ F/m | 21-22 2(c), 20-21 3(c) |
| | Stress in the dielectric | $g = \frac{Q}{2\pi\varepsilon_0\varepsilon_r\, x} = \frac{V}{x\,\ln(D/d)}$ | 21-22 2(c) |
| | Cable charging current | $I_c = 2\pi f C V$ | 21-22 2(c) |
| Capacitance (10) | **Belted cable tests** | $C_1 = 3C_s,\ C_2 = C_s + 2C_c$ | 20-21 3(d) |
| | **Capacitance per phase** | $C_N = C_s + 3C_c$ | 20-21 3(d) |
| | **Cable charging current** | $I_c = 2\pi f\, C_N\, V_{ph}$ | 20-21 3(d) |
| Grading (20) | Layer stresses | $g_{1max} = \frac{Q}{\pi\varepsilon_0\varepsilon_1 d},\ g_{2max} = \frac{Q}{\pi\varepsilon_0\varepsilon_2 d_1}$ | 21-22 4(d) |
| | Two dielectrics, stresses | $\varepsilon_1 d\, g_{1max} = \varepsilon_2 d_1\, g_{2max}$ | 21-22 4(d) |
| | Two dielectrics, voltage | $V = \frac{g_{1max}\,d}{2}\ln\frac{d_1}{d} + \frac{g_{2max}\,d_1}{2}\ln\frac{D}{d_1}$ | 21-22 4(d) |
| | **Capacitance grading** | $V = \frac{g_{1max}\,d}{2}\ln\frac{d_1}{d} + \frac{g_{2max}\,d_1}{2}\ln\frac{d_2}{d_1} + \frac{g_{3max}\,d_2}{2}\ln\frac{D}{d_2}$ | 21-22 2(d) |
| | Equal maximum stress | $\varepsilon_1 d = \varepsilon_2 d_1 = \varepsilon_3 d_2$ | 21-22 2(d) |
| | Graded voltage | $V = \frac{g_{max}}{2}\left[d\ln\frac{d_1}{d} + d_1\ln\frac{d_2}{d_1} + d_2\ln\frac{D}{d_2}\right]$ | 21-22 2(d) |
| Insulation resistance (5) | **Insulation resistance** | $R = \frac{\rho}{2\pi l}\ln\frac{r_2}{r_1}$ | 21-22 4(b) |
| Economical size (2) | **Most economical size** | $d = \frac{2V}{g_{max}},\ D = e\,d = 2.718\,d$ | 21-22 6(a) |

## Symbols:

$Q$ = charge on the core (C/m) ↩

$x$ = distance from the centre (m) ↩

$\varepsilon_r$ = relative permittivity ↩

$d$ = core diameter ↩

$D$ = inside diameter of the sheath ↩

$V$ = peak core-to-sheath voltage (kV) ↩

$C_s$ = core to sheath ↩

$C_c$ = core to core ↩

$d_1$ = outer diameter of layer 1 (cm) ↩

$\varepsilon_1$, $\varepsilon_2$ = relative permittivities of the inner and outer layers ↩

$d_2$ = outer diameter of layer 2 (cm) ↩

$g_{max}$ = common maximum stress (kV/cm) ↩

$\rho$ = resistivity ($\Omega \cdot$ cm) ↩

$l$ = length (cm) ↩

$r_1$, $r_2$ = core and sheath radii ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Dielectric stress** | | | | |
| 1 | 21-22 2(c) | 8 | numerical | A single core cable for use on 11 kV, 50 Hz system has conductor area of 0.645 $cm^2$ and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find: (i) the maximum electrostatic stress in the cable, (ii) minimum electrostatic stress in the cable, (iii) capacitance of the cable per km length, and (iv) charging current. |
| 2 | 21-22 TT2 8(b) | 5 | numerical repeats 21-22 2(c) | A single core cable for use on 11 kV, 50 Hz system has a conductor area of 0.645 $cm^2$ and internal diameter of sheath is 2.18 cm. The permittivity of the dielectric used in the cable is 3.5. Find (i) the maximum electrostatic stress in the cable (ii) the minimum electrostatic stress in the cable (iii) the capacitance of the cable per km length (iv) the charging current. |
| 3 | 20-21 2(d) | 10 | numerical | A 33 kV single core cable has a conductor diameter of 1 cm and a sheath of inside diameter 4 cm. Find the maximum and minimum stress in the insulation. |
| 4 | 20-21 3(c) | 10 | derivation | Derive the expression of dielectric stress in a single core cable. |
| 5 | 20-21 4(c) | 10 | numerical | The maximum and minimum stresses in the dielectric of a single core cable are 40 kV/cm (r.m.s.) and 10 kV/cm (r.m.s.) respectively. If the conductor diameter is 2 cm, find : (i) thickness of insulation (ii) operating voltage |
| **Capacitance** | | | | |
| 6 | 20-21 3(d) | 10 | numerical | The capacitances of a 3-phase belted cable are $12 \cdot 6$ $\mu$F between the three cores bunched together and the lead sheath and $7 \cdot 4$ $\mu$F between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply. |
| **Construction** | | | | |
| 7 | 21-22 2(b) | 5 | theory | Write down the general construction of a 3-conductor cable with required figure. |
| **Grading** | | | | |
| 8 | 21-22 2(d) | 10 | derivation | Define grading of cables. Why is unequal stress distribution in a cable undesirable? Consider three dielectric layers for capacitance grading and derive the expression of potential difference between core and earthed sheath. |
| 9 | 21-22 4(d) | 10 | numerical | A 66 kV single-core lead sheathed cable is graded by using two dielectrics of relative permittivity 5 and 3 respectively; thickness of each being 1 cm. The core diameter is 2 cm. Determine the maximum stress in the two dielectrics. |
| **Insulation resistance** | | | | |
| 10 | 21-22 4(b) | 5 | numerical | The insulation resistance of a single-core cable is 495 MΩ per km. If the core diameter is 2.5 cm and the resistivity of insulation is $4.5 \times 10^{14}$ $\Omega - cm$, find the insulation thickness. |
| **Economical size** | | | | |
| 11 | 21-22 6(a) | 2 | numerical | Find the most economical size of a single-core cable working on a 132 kV, 3-phase system, if a dielectric stress of 60 kV/cm can be allowed. |


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## 4.7. Cable faults:

20 marks, 3 questions, in 2 of 2 papers. Book: Mehta Ch. 11.

Most often numerical (3 of 3). Subtopics: Loop tests.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Loop tests (20) | **Varley loop balance** | $\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X + S}$, $R_X = \frac{Q\,R_{loop} - P\,S}{P+Q}$ | 20-21 5(b) |
| | Equal ratio arms | $R_X = \frac{R_{loop} - S}{2}$ | 20-21 5(b) |
| | **Murray loop balance** | $\frac{P}{Q} = \frac{R_{loop} - R_X}{R_X}$, $R_X = \frac{Q}{P+Q} R_{loop}$ | 21-22 1(b) |

### Symbols:

$S$ = variable resistance in series with the faulty conductor ($\Omega$) ↩

$R_{loop}$ = faulty + healthy conductor ($\Omega$); faulty + sound core ($\Omega$) ↩

$P$, $Q$ = ratio arms ($\Omega$) ↩

$R_X$ = resistance up to the fault ($\Omega$) ↩

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Loop tests** | | | | |
| 1 | 21-22 1(b) | 5 | numerical | In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio of 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. |
| 2 | 21-22 TT3 9(a) | 5 | numerical repeats 21-22 1(b) | In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000 m. The resistance of the other two arms of the testing network at balance are in the ratio 2.75 : 1. Calculate the distance of the fault from the testing end of the cable. |
| 3 | 20-21 5(b) | 10 | numerical | Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 ohm. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 ohm. The fixed resistors have equal values. Calculate the distance of the fault from the test end. |

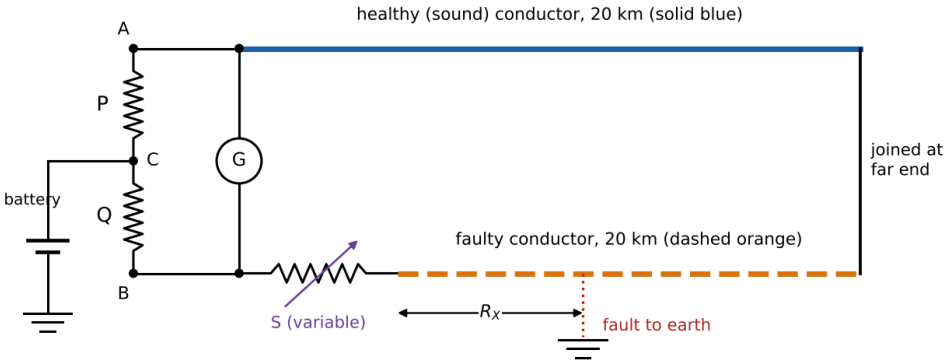



**Figure 5:** 20-21 5(b): the Varley loop with S in series with the faulty conductor

## 4.8. Power system stability:

43 marks, 5 questions, in 2 of 2 papers. Book: Grainger Ch. 16.

Most often theory (3 of 5). Subtopics: Equal area criterion; Transient EMF.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| Equal area criterion (23) | **Power-angle equation** | $P_e = \dfrac{\lvert E\rvert\lvert V\rvert}{X}\sin\delta = P_{max}\sin\delta$ | 20-21 6(a) |
| | Swing equation | $M\dfrac{d^2\delta}{dt^2} = P_m - P_e$ | 21-22 6(c), 20-21 6(a) |
| | **Equal area criterion** | $\displaystyle\int_{\delta_0}^{\delta_{max}} (P_m - P_e)\,d\delta = 0 \;\Leftrightarrow\; A_1 = A_2$ | 20-21 6(a) |
| | Critical clearing angle | $\cos\delta_{cr} = \dfrac{P_m(\delta_{max}-\delta_0) + P_{3max}\cos\delta_{max} - P_{2max}\cos\delta_0}{P_{3max} - P_{2max}}$ | 20-21 6(a) |
| | Limit of the swing | $\delta_{max} = 180° - \delta_1$ | 20-21 6(b) |
| | **Equal area criterion** | $A_1 = A_2$ | 21-22 6(c) |
| Transient EMF (20) | Terminal voltage angle | $\sin\theta = \dfrac{P\,X}{\lvert V_t\rvert\lvert V\rvert}$ | 20-21 6(c) |
| | Current | $I = \dfrac{V_t - V}{jX}$ | 20-21 6(c) |
| | **Transient emf** | $E' = V_t + jX'_dI$ | 21-22 6(d), 20-21 6(c) |
| | **Maximum power** | $P_{max} = \dfrac{\lvert E'\rvert\lvert V\rvert}{X}$ | 20-21 6(c) |
| | Transfer reactance (Y to Δ) | $X_{ab} = \dfrac{X_aX_b + X_bX_c + X_cX_a}{X_c}$ | 21-22 6(d), 20-21 6(c) |
| | **Power-angle equation** | $P_e = \dfrac{\lvert E'\rvert\lvert V\rvert}{X}\sin\delta = P_{max}\sin\delta$ | 21-22 6(d) |


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| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|

## Symbols:

$\delta$ = power angle (E ahead of V); rotor angle ↩

$M$ = inertia constant ↩

$P_m$ = mechanical input (pu) ↩

$A_1$ = accelerating area

$A_2$ = decelerating area

$\delta_1$ = new operating angle ↩

$A_1$, $A_2$ = accelerating and decelerating areas ↩

$X$ = reactance from $V_t$ to the bus (pu); transfer reactance (pu) ↩

$X'_d$ = transient reactance (pu) ↩

$V_t$ = terminal voltage (pu) ↩

$I$ = current (pu) ↩

$X_c$ = branch to the faulted node (pu) ↩

$E'$ = transient emf (pu) ↩

$V$ = infinite-bus voltage (pu) ↩

## Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **Equal area criterion** | | | | |
| 1 | 21-22 6(c) | 8 | theory repeats 20-21 6(b) | Explain what happens for sudden increase in mechanical input to generator using $Pe - \delta$ diagram. |
| 2 | 20-21 6(a) | 5 | theory | Define Power Angle. Briefly explain Equal Area Criterion. |
| 3 | 20-21 6(b) | 10 | theory | Explain what happens for sudden increase in mechanical input to generator using Pe-$\delta$ diagram. |
| **Transient EMF** | | | | |
| 4 | 21-22 6(d) | 10 | numerical repeats 20-21 6(c) | The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus ($\|V\| = 1.0\ pu$), with the generator terminal voltage ($\|V_t\| = 1.0\ pu$). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (i) System healthy, (ii) One line shorted (3 phase) in the middle and (iii) One line open. Plot all the three power angle curves. |
| 5 | 20-21 6(c) | 10 | numerical | The generator of Fig. 2 is delivering 1.0 pu power to the infinite bus (\|V\|=1.0 pu), with the generator terminal voltage (\|Vt\|=1.0 pu). Calculate the generator emf behind transient reactance. Find the maximum power that can be transferred under the following conditions: (a) System healthy (b) One line shorted (3 phase) in the middle. |

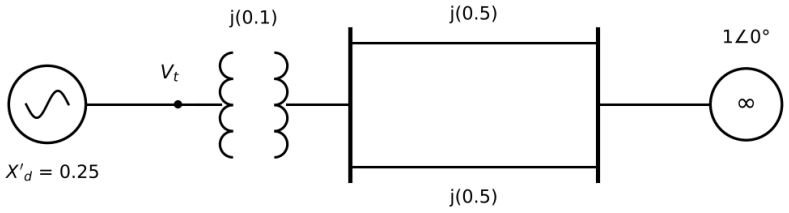



**Figure 6:** 21-22 6(d): Fig. 2 of the paper (redrawn): figure for Q-6(d)


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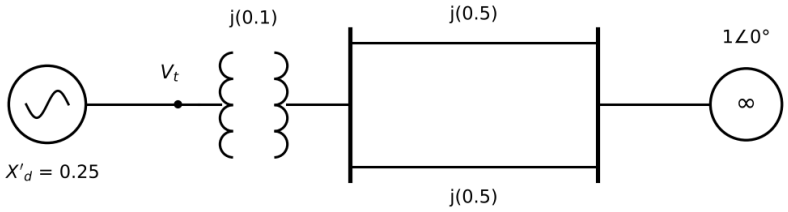



**Figure 7:** 20-21 6(c): Fig. 2 of the paper (redrawn)

## 4.9. HVDC and FACTS:

11 marks, 3 questions, in 2 of 2 papers. Book: Hingorani and Gyugyi; Rashid Ch. 12.

Most often theory (3 of 3). Subtopics: FACTS; HVDC.

### Formulas:

| Subtopic | Formula | Equation | Used in |
|---|---|---|---|
| FACTS (4) | **Power over a line** | $P = \dfrac{V_1 V_2}{X} \sin\delta$ | 20-21 3(b) |
| HVDC (7) | Annual cost | $P_1 + P_2 a + \dfrac{P_3}{a}$ | 21-22 6(b) |
| | **Kelvin's law** | $P_2 a = \dfrac{P_3}{a}$ | 21-22 6(b) |

### Symbols:

$a$ = conductor area ↩ $P_2 a$ = interest and depreciation ↩ $P_3/a$ = cost of energy lost

### Questions:

| No. | Paper | Marks | Type | Question |
|---|---|---|---|---|
| **FACTS** | | | | |
| 1 | 20-21 3(b) | 4 | theory | Describe different FACTS devices and applications. |
| **HVDC** | | | | |
| 2 | 21-22 2(a) | 2 | theory | What is monopolar link? |
| 3 | 21-22 6(b) | 5 | theory | Define HVDC. Discuss the Economics of Power transmission. |

# 5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| **Line inductance** | | | | |
| 1 | 20-21 1(a) | 1 | Basics | Inductance is flux linkage per ampere, $L = \psi/I = N\phi/I$ henry; it opposes a change of current, $e = L\,di/dt$; for a line, $L = 2 \times 10^{-7} \ln(D/r')$ H/m per conductor. |
| 2 | 20-21 2(b) | 4 | Transposition | A transposed line exchanges conductor positions at intervals so each occupies each position for one third of the length. Reason: with unsymmetrical spacing the phases have unequal $L$ and $C$; transposition equalises them, keeps the voltage drops balanced and reduces telephone interference. |

| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| 3 | 20-21 3(a) | 1 | Skin effect | Skin effect: alternating current crowds towards the surface because the inner filaments link more flux and have more reactance; the effective area falls, so the ac resistance exceeds the dc resistance; it grows with frequency, size and conductivity; stranded and hollow conductors reduce it. |
| **Line capacitance** | | | | |
| 4 | 21-22 5(a) | 2 | Effect of earth | The earth is a zero-potential conductor; by the method of images it acts as equal and opposite charges at the same depth below ground; this raises the line capacitance slightly ($C = \pi\varepsilon_0 / \ln[(d/r) \cdot 2h/\sqrt{4h^2 + d^2}]$ for a single-phase line); small for high lines. |
| **Corona** | | | | |
| 5 | 21-22 1(a) | 2 | Critical disruptive voltage | $V_c$: the least phase-to-neutral voltage at which the surface field reaches the breakdown strength of air (30 kV/cm peak) and corona starts, $V_c = m_0 g_0 \delta r \ln(d/r)$ kV; below it no corona; above it corona loss, interference and ozone; lines are designed below it. |
| 6 | 20-21 2(a) | 1 | Corona effect | Corona: violet glow, hissing, ozone and power loss round a conductor when the surface field exceeds the strength of air (30 kV/cm peak) and the air ionises. Depends on voltage, conductor size and surface, spacing and weather. |
| 7 | 20-21 4(a) | 5 | Critical disruptive voltage | $V_c$: the least phase-to-neutral voltage at which the surface field reaches the strength of air (30 kV/cm peak) and corona starts: $V_c = m_0 g_0 \delta r \ln(d/r)$ kV. Significance: above it corona loss, interference and ozone appear; lines are designed to work below it. |
| **Overhead insulators** | | | | |
| 8 | 21-22 3(a) | 2 | Failure | By flash-over (an arc through the air from the conductor to the pin; the insulator survives) or puncture (a discharge through the porcelain; destroyed). Safety factor = puncture strength / flash-over voltage, about 10, so flash-over comes first. |
| 9 | 20-21 4(b) | 10 | String efficiency | $\eta = V/(nV_n)$; the shunt capacitance $KC$ makes the line unit take the most voltage, so $\eta < 100$ %. Improve it with longer cross-arms (smaller $K$), capacitance grading of the discs, and a guard (static shielding) ring at line potential. |
| 10 | 20-21 5(a) | 5 | Failure | Cracking from unequal expansion of porcelain, steel and cement; impure material; porosity; poor glazing; flash-over (arc heat cracks it); mechanical stress (tension, wind, ice); short circuits. |
| **Mechanical design** | | | | |
| 11 | 21-22 3(b) | 5 | Supports | Wooden poles (cheap, spans of 40 to 50 m, rural), steel poles (rail or tubular, 50 to 80 m, towns), RCC poles (strong, long life, 80 to 200 m); A- or H-type double poles for heavy loads. |
| 12 | 21-22 4(a) | 2 | Sag | A stringing chart: curves of sag and tension against temperature for a given conductor and span; the lineman reads the right sag (or tension) for the temperature at erection, so the line is safe at both temperature extremes. |
| **Underground cables** | | | | |
| 13 | 21-22 2(b) | 5 | Construction | From the centre: three stranded cores, each paper (or PVC/XLPE) insulated; jute fillers; a common paper belt; lead sheath; jute bedding; steel-wire or tape armour; jute serving. |
| **Power system stability** | | | | |
| 14 | 21-22 6(c) | 8 | Equal area criterion | A sudden rise of $P_m$ makes $P_m > P_e$: the rotor accelerates, $\delta$ passes the new equilibrium $\delta_1$ (area $A_1$), overshoots to $\delta_2$ where the decelerating area $A_2 = A_1$, then swings back and, with damping, settles at $\delta_1$; unstable if $A_1$ exceeds the area available up to $180° - \delta_1$. |


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| No. | Paper | Marks | Subtopic | Points to cover |
|---|---|---|---|---|
| 15 | 20-21 6(a) | 5 | Equal area criterion | Power angle $\delta$: the angle between the internal emf and the infinite-bus voltage; $P_e = (EV/X)\sin\delta$. Equal area criterion: after a disturbance the machine stays in step if the decelerating area ($P_e > P_m$) can equal the accelerating area ($P_m > P_e$). |
| 16 | 20-21 6(b) | 10 | Equal area criterion | A sudden rise of $P_m$ makes $P_m > P_e$: the rotor accelerates, $\delta$ passes the new equilibrium $\delta_1$ (area $A_1$), overshoots to $\delta_2$ where the decelerating area $A_2 = A_1$, then swings back and, with damping, settles at $\delta_1$; unstable if $A_1$ exceeds the area available up to $180° - \delta_1$. |
| **HVDC and FACTS** | | | | |
| 17 | 21-22 2(a) | 2 | HVDC | Monopolar HVDC link: one conductor (usually negative) between the converter stations, earth or sea as the return; cheap, used for submarine cables; bipolar and homopolar are the other kinds. |
| 18 | 21-22 6(b) | 5 | HVDC | HVDC: power as high-voltage dc between a rectifier and an inverter station. Economics: Kelvin's law for conductor size (annual loss cost = annual capital charge on the area-dependent part), an economic voltage, and HVDC cheaper than HVAC beyond the break-even distance (lines and cables) despite costly terminals. |
| 19 | 20-21 3(b) | 4 | FACTS | FACTS controllers change $V$, $X$ or $\delta$ in $P = V_1V_2\sin\delta/X$. Shunt (SVC, STATCOM): voltage control and reactive support. Series (TCSC, SSSC): lower $X$, more power, flow control, damping. Combined (UPFC, IPFC): $V$, $X$ and $\delta$ together. |

# 6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

| No. | Paper | Marks | Derivation | Outline |
|---|---|---|---|---|
| **Line inductance** | | | | |
| 1 | 21-22 1(c) | 8 | Transposition | Flux of a line current $2\times10^{-7}I\ln(1/D)$; $\psi_a$ in the three sections; average: the distances multiply; $I_b + I_c = -I_a$; $\psi_a = 2\times10^{-7}I_a\ln(D_{eq}/r')$; $L = \psi_a/I_a$. |
| 2 | 21-22 4(c) | 8 | Composite conductors | Flux of a line current $2\times10^{-7}I\ln(1/D)$, with $r'$ for itself; flux of strand $a$; combine the logs; $L_a = \psi_a/(I/n)$; every strand; $n$ strands in parallel, $L_X = \sum L/n^2$; mutual GMD over self GMD; loop $L = L_X + L_Y$. |
| 3 | 20-21 2(c) | 10 | Composite conductors | Flux of a line current $2\times10^{-7}I\ln(1/D)$, with $r'$ for itself; flux of strand $a$; combine the logs; $L_a = \psi_a/(I/n)$; every strand; $n$ strands in parallel, $L_X = \sum L/n^2$; mutual GMD over self GMD; loop $L = L_X + L_Y$. |
| **Line capacitance** | | | | |
| 4 | 21-22 5(d) | 10 | Unsymmetrical spacing | Line charge: $V = (q/2\pi\varepsilon_0)\ln(D_2/D_1)$; $V_{ab}$ in the three transposition sections; average: the distances multiply, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$; same for $V_{ac}$; add with $q_a + q_b + q_c = 0$; $V_{ab} + V_{ac} = 3V_{an}$; $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$. |
| 5 | 21-22 TT1 7(b) | 0 | Unsymmetrical spacing | Line charge: $V = (q/2\pi\varepsilon_0)\ln(D_2/D_1)$; $V_{ab}$ in the three transposition sections; average: the distances multiply, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$; same for $V_{ac}$; add with $q_a + q_b + q_c = 0$; $V_{ab} + V_{ac} = 3V_{an}$; $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$. |
| 6 | 20-21 1(c) | 10 | Unsymmetrical spacing | Line charge: $V = (q/2\pi\varepsilon_0)\ln(D_2/D_1)$; $V_{ab}$ in the three transposition sections; average: the distances multiply, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$; same for $V_{ac}$; add with $q_a + q_b + q_c = 0$; $V_{ab} + V_{ac} = 3V_{an}$; $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$. |
| **Overhead insulators** | | | | |
| 7 | 21-22 3(d) | 10 | String efficiency | Each capacitance carries $\omega CV$; KCL at A: $V_2 = (1+K)V_1$; KCL at B: $V_3 = (1+3K+K^2)V_1$; add for $V$; $\eta = V/(3V_3)$; evaluate at $K = 0.2$, 0.8 and for more discs. |

| No. | Paper | Marks | Derivation | Outline |
|---|---|---|---|---|
| **Underground cables** | | | | |
| 8 | 21-22 2(d) | 10 | Grading | Gauss: $D_x = Q/2\pi x$ in every layer; $g_k = Q/(2\pi\varepsilon_0\varepsilon_k x)$; maximum at each inner surface; integrate each layer; add the three; equal maxima need $\varepsilon_1 d = \varepsilon_2 d_1 = \varepsilon_3 d_2$. |
| 9 | 20-21 3(c) | 10 | Dielectric stress | Gauss: $D_x = Q/2\pi x$; $g_x = Q/(2\pi\varepsilon_0\varepsilon_r x)$; integrate from $d/2$ to $D/2$: $V = (Q/2\pi\varepsilon_0\varepsilon_r)\ln(D/d)$; remove $Q$: $g_x = V/(x\ln(D/d))$; $g_{max}$ at the core, $g_{min}$ at the sheath. |

# 7. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

| No. | Paper | Marks | Subtopic | Method | Answer |
|---|---|---|---|---|---|
| **Line inductance** | | | | | |
| 1 | 21-22 1(d) | 10 | Mutual inductance | $D_1$ and $D_2$ from the geometry of Fig. 1; $M = 4\times10^{-7}\ln(D_2/D_1)$ (both power wires add); $E = \omega M I$. | $M = 0.0758$ mH/km; 2.38 V/km |
| 2 | 21-22 TT1 7(a) | 5 | Bundled conductors | $r' = 0.7788r$; bundle $D_s = \sqrt{r'd}$; $D_{eq}$ from the bundle centres; $L = 2\times10^{-7}\ln(D_{eq}/D_s)$; $X_L = 2\pi f L$. | $L = 1.102$ mH/km; $X_L = 0.346\ \Omega$/km |
| 3 | 21-22 TT3 9(b) | 5 | Mutual inductance | Distances from each power wire to each telephone wire; flux of $P_1$ and $P_2$ linking the telephone loop; $M = 2\times10^{-7}\ln[(D_{P_1T_2}D_{P_2T_1})/(D_{P_1T_1}D_{P_2T_2})]$; $E = \omega M I$. | $M = 0.648\ \mu$H/km; 30.6 mV/km |
| 4 | 20-21 1(b) | 4 | Bundled conductors | Bundle GMR $D_s^b = \sqrt{D_s d}$ with the table $D_s$; $D_{eq} = \sqrt[3]{D_{ab}D_{bc}D_{ca}}$; $L = 2\times10^{-7}\ln(D_{eq}/D_s^b)$; $X_L = 2\pi f L$ at 50 Hz (handwritten on the scan). | $X_L = 0.304\ \Omega$/km $= 0.489\ \Omega$/mi per phase (0.365 $\Omega$/km at 60 Hz) |
| 5 | 20-21 1(d) | 10 | Mutual inductance | Distances from each power wire to each telephone wire; flux of $P_1$ and $P_2$ linking the telephone loop; $M = 2\times10^{-7}\ln[(D_{P_1T_2}D_{P_2T_1})/(D_{P_1T_1}D_{P_2T_2})]$; $E = \omega M I$. | $M = 0.648\ \mu$H/km; 30.6 mV/km |
| **Line capacitance** | | | | | |
| 6 | 21-22 5(b) | 5 | Unsymmetrical spacing | $D_{eq} = \sqrt[3]{D\cdot D\cdot 2D}$; $C_n = 2\pi\varepsilon_0/\ln(D_{eq}/r)$; $I_c = \omega C_n V_{ph}$. | $C_n = 0.00826\ \mu$F/km; 0.165 A/km |
| **Overhead insulators** | | | | | |
| 7 | 21-22 3(c) | 8 | String efficiency | $V = 33/\sqrt{3}$ kV; $V_2 = (1+K)V_1$, $V_3 = (1+3K+K^2)V_1$; $V_1$ from $V = V_1+V_2+V_3$; $\eta = V/(3V_3)$. | 5.52, 6.13, 7.41 kV; $\eta = 85.74$ % |
| 8 | 21-22 5(c) | 8 | String efficiency | $V_2 = (1+K)V_1$ gives $K$; $V_3 = V_2 + K(V_1+V_2)$; phase voltage $= V_1+V_2+V_3$; line voltage $\sqrt{3}$ times; $\eta = V/(3V_3)$. | $K = 0.375$; $V_L = 64.3$ kV; $\eta = 68.28$ % |
| 9 | 21-22 TT2 8(a) | 5 | String efficiency | $V = 100/\sqrt{3}$ kV; $V_{k+1} = V_k + K(V_1+\cdots+V_k)$ for five discs; $V_1$ from the sum; $\eta = V/(5V_5)$. | 8.00, 8.80, 10.48, 13.21, 17.25 kV; $\eta = 66.92$ % |
| **Mechanical design** | | | | | |
| 10 | 20-21 5(c) | 10 | Sag | Working tension $T$ = ultimate strength / safety factor; sag $S = wl^2/(8T)$; support height = clearance + sag. | sag 3.71 m; supports at 13.71 m |

| No. | Paper | Marks | Subtopic | Method | Answer |
|---|---|---|---|---|---|
| **Underground cables** | | | | | |
| 11 | 21-22 2(c) | 8 | Dielectric stress | $g_{max} = 2V/(d\ln(D/d))$ at the core, $g_{min} = 2V/(D\ln(D/d))$ at the sheath; $C = 2\pi\varepsilon_0\varepsilon_r/\ln(D/d)$; $I_c = \omega CV$. | $g_{max} = 27.66$, $g_{min} = 11.50$ kV/cm; $C = 0.2218$ $\mu$F/km; $I_c = 0.767$ A/km |
| 12 | 21-22 4(b) | 5 | Insulation resistance | $R = (\rho/2\pi l)\ln(r_2/r_1)$; solve $\ln(r_2/r_1) = 2\pi lR/\rho$; thickness $r_2 - r_1$. | 1.245 cm |
| 13 | 21-22 4(d) | 10 | Grading | The same $Q$ passes both layers, so $g_{2max}/g_{1max} = \varepsilon_1 d/(\varepsilon_2 d_1)$; $V = (g_{1max}d/2)\ln(d_1/d) + (g_{2max}d_1/2)\ln(D/d_1)$; solve for $g_{1max}$. | 48.21 kV/cm (inner, $\varepsilon_r = 5$); 40.18 kV/cm (outer) r.m.s. |
| 14 | 21-22 6(a) | 2 | Economical size | Peak phase voltage $V = 132\sqrt{2}/\sqrt{3}$; most economical core $d = 2V/g_{max}$ (from $D/d = e$); sheath $D = ed$. | $d = 3.59$ cm; $D = 9.77$ cm |
| 15 | 21-22 TT2 8(b) | 5 | Dielectric stress | $g_{max} = 2V/(d\ln(D/d))$ at the core, $g_{min} = 2V/(D\ln(D/d))$ at the sheath; $C = 2\pi\varepsilon_0\varepsilon_r/\ln(D/d)$; $I_c = \omega CV$. | $g_{max} = 27.66$, $g_{min} = 11.50$ kV/cm; $C = 0.2218$ $\mu$F/km; $I_c = 0.767$ A/km |
| 16 | 20-21 2(d) | 10 | Dielectric stress | $g_{max} = 2V/(d\ln(D/d))$, $g_{min} = 2V/(D\ln(D/d))$ with $V = 33$ kV core to sheath. | $g_{max} = 47.61$ kV/cm, $g_{min} = 11.90$ kV/cm (r.m.s.) |
| 17 | 20-21 3(d) | 10 | Capacitance | Three cores bunched to sheath: $3C_s$ measured; one core to the other two and sheath: $C_s + 2C_c$; per-phase $C_N = C_s + 3C_c$; $I_c = \omega C_N V_{ph}$. | $C_s = 4.2$, $C_c = 1.6$, $C_N = 9.0$ $\mu$F; $I_c = 107.7$ A |
| 18 | 20-21 4(c) | 10 | Dielectric stress | $g_{max}/g_{min} = D/d$ gives $D$; thickness $(D-d)/2$; $V = g_{max}(d/2)\ln(D/d)$. | thickness 3 cm ($D = 8$ cm); $V = 55.45$ kV r.m.s. |
| **Cable faults** | | | | | |
| 19 | 21-22 1(b) | 5 | Loop tests | Loop resistance $R_{loop}$ = faulty + sound cable; at balance $R_X/(R_{loop} - R_X) = Q/P$, so $R_X = R_{loop}/(1 + P/Q)$; distance = $R_X$ / resistance per metre. | fault at 433.3 m from the testing end |
| 20 | 21-22 TT3 9(a) | 5 | Loop tests | Loop resistance $R_{loop}$ = faulty + sound cable; at balance $R_X/(R_{loop} - R_X) = Q/P$, so $R_X = R_{loop}/(1 + P/Q)$; distance = $R_X$ / resistance per metre. | fault at 433.3 m from the testing end |
| 21 | 20-21 5(b) | 10 | Loop tests | Varley loop: $R_{loop}$ = twice the cable resistance; with $P = Q$, $R_X = (R_{loop} - S)/2$; distance = $R_X$ / resistance per km. | 15 km from the test end |
| **Power system stability** | | | | | |
| 22 | 21-22 6(d) | 10 | Transient EMF | $V_t$ angle from $P = V_t V\sin\theta/X$; current $I = (V_t - V)/jX$; $E' = V_t + jX'_dI$; $P_{max} = E'V/X_{transfer}$ for each network (fault: star-delta or Thevenin reduction). | $E' = 1.0747\angle 33.94°$ pu; $P_{max} = 1.791$ (healthy), 0.693 (mid-line fault), 1.264 pu (one line open) |
| 23 | 20-21 6(c) | 10 | Transient EMF | $V_t$ angle from $P = V_t V\sin\theta/X$; current $I = (V_t - V)/jX$; $E' = V_t + jX'_dI$; $P_{max} = E'V/X_{transfer}$ for each network (fault: star-delta or Thevenin reduction). | $E' = 1.0747\angle 33.94°$ pu; $P_{max} = 1.791$ pu healthy, 0.693 pu with a mid-line fault |


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# 8. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

| No. | Paper | Kind | Diagram |
|---|---|---|---|
| **Line inductance** | | | |
| 1 | 21-22 1(c) | Diagram | flux of one conductor |
| 2 | 21-22 1(c) | Diagram | one transposition cycle |
| 3 | 21-22 1(d) | Diagram | flux of the power wires linking the telephone loop |
| 4 | 21-22 4(c) | Diagram | composite conductors X ($n$ strands) and Y ($m$ strands) |
| 5 | 21-22 TT1 7(a) | Diagram | two-conductor bundle and its GMR |
| 6 | 21-22 TT3 9(b) | Diagram | power and telephone lines in one plane |
| 7 | 20-21 1(a) | Diagram | flux of one current-carrying conductor, inside and outside |
| 8 | 20-21 1(b) | Diagram | two-conductor bundle and its GMR |
| 9 | 20-21 1(d) | Diagram | power and telephone lines in one plane |
| 10 | 20-21 2(b) | Diagram | one transposition cycle |
| 11 | 20-21 2(c) | Diagram | composite conductors X ($n$ strands) and Y ($m$ strands) |
| 12 | 20-21 3(a) | Diagram | current density across a conductor, dc and ac |
| **Line capacitance** | | | |
| 13 | 21-22 5(a) | Diagram | two-wire line above the earth with image charges |
| 14 | 21-22 5(b) | Diagram | three conductors in a horizontal row |
| 15 | 21-22 5(d) | Diagram | three charged conductors at unequal spacing |
| 16 | 21-22 5(d) | Diagram | one transposition cycle |
| 17 | 21-22 TT1 7(b) | Diagram | three charged conductors at unequal spacing |
| 18 | 21-22 TT1 7(b) | Diagram | one transposition cycle |
| 19 | 20-21 1(c) | Diagram | three charged conductors at unequal spacing |
| 20 | 20-21 1(c) | Diagram | one transposition cycle |
| **Corona** | | | |
| 21 | 21-22 1(a) | Diagram | field lines and the corona shell round a conductor |
| 22 | 20-21 2(a) | Diagram | field lines and the ionised shell round a conductor |
| **Overhead insulators** | | | |
| 23 | 21-22 3(a) | Diagram | flash-over and puncture paths of a pin insulator |
| 24 | 21-22 3(c) | Circuit | equivalent circuit of a three-disc string |
| 25 | 21-22 3(d) | Graph | voltage share per disc for several $K$ |
| 26 | 21-22 3(d) | Graph | efficiency against the number of discs |
| 27 | 21-22 5(c) | Circuit | equivalent circuit of the three-disc string with the measured voltages |
| 28 | 21-22 TT2 8(a) | Graph | voltage on each disc of the five-disc string |
| 29 | 20-21 4(b) | Circuit | equivalent circuit of a three-disc string |
| 30 | 20-21 4(b) | Diagram | three ways of improving string efficiency |
| 31 | 20-21 4(b) | Graph | efficiency against the number of discs |
| 32 | 20-21 5(a) | Diagram | flash-over and puncture paths of a pin insulator |
| **Mechanical design** | | | |
| 33 | 21-22 3(b) | Diagram | wooden, steel and RCC poles |
| 34 | 21-22 4(a) | Graph | sag and tension against temperature |
| 35 | 20-21 5(c) | Diagram | conductor between supports with sag and clearance |
| **Underground cables** | | | |
| 36 | 21-22 2(b) | Diagram | cross-section of a three-core belted cable |


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| No. | Paper | Kind | Diagram |
|---|---|---|---|
| 37 | 21-22 2(c) | Graph | stress across the insulation, largest at the core |
| 38 | 21-22 2(d) | Diagram | cable with three dielectric layers and the stress profile |
| 39 | 21-22 4(b) | Diagram | radial leakage current in the insulation |
| 40 | 21-22 4(d) | Graph | stress across the graded and ungraded cable |
| 41 | 21-22 6(a) | Graph | sheath diameter against core diameter |
| 42 | 21-22 TT2 8(b) | Graph | stress across the insulation, largest at the core |
| 43 | 20-21 2(d) | Graph | stress profile of this cable |
| 44 | 20-21 3(c) | Diagram | cable section with the Gauss cylinder |
| 45 | 20-21 3(d) | Diagram | belted cable construction |
| 46 | 20-21 3(d) | Diagram | core-to-core and core-to-sheath capacitances |
| 47 | 20-21 4(c) | Graph | stress across this cable |
| **Cable faults** | | | |
| 48 | 21-22 1(b) | Circuit | Murray loop test |
| 49 | 21-22 TT3 9(a) | Circuit | Murray loop test |
| 50 | 20-21 5(b) | Circuit | Varley loop test |
| **Power system stability** | | | |
| 51 | 21-22 6(c) | Graph | $P_e$–$\delta$ curve with the accelerating and decelerating areas |
| 52 | 21-22 6(c) | Graph | rotor angle swing after the step |
| 53 | 21-22 6(d) | Diagram | network with the mid-line fault and its reduction |
| 54 | 21-22 6(d) | Graph | power-angle curves for the three cases |
| 55 | 20-21 6(a) | Diagram | phasor diagram with the power angle |
| 56 | 20-21 6(a) | Graph | equal area criterion for a cleared fault |
| 57 | 20-21 6(b) | Graph | $P_e$–$\delta$ curve with the accelerating and decelerating areas |
| 58 | 20-21 6(b) | Graph | rotor angle swing after the step |
| 59 | 20-21 6(c) | Diagram | network with the mid-line fault and its reduction |
| 60 | 20-21 6(c) | Graph | power-angle curves |
| **HVDC and FACTS** | | | |
| 61 | 21-22 2(a) | Diagram | monopolar, bipolar and homopolar HVDC links |
| 62 | 21-22 6(b) | Graph | Kelvin's law, capital charge and loss cost against area |
| 63 | 21-22 6(b) | Graph | total cost against distance for HVAC and HVDC |
| 64 | 20-21 3(b) | Diagram | shunt, series and combined FACTS connections |

# 9. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

| Pair | Key difference | Asked in |
|---|---|---|
| **Corona** | | |
| Corona vs skin effect | air ionises round a high-voltage conductor vs ac current crowds to the conductor surface | 20-21 2(a) |
| Critical disruptive vs visual critical voltage | corona starts (no glow yet) vs the glow becomes visible, higher | 20-21 4(a) |
| **Overhead insulators** | | |
| Flash-over vs puncture | arc through the air outside, insulator survives vs discharge through the porcelain, insulator destroyed | 21-22 3(a), 20-21 5(a) |
| **Underground cables** | | |
| Capacitance vs intersheath grading | layers of different permittivity vs metal intersheaths held at set voltages | 21-22 2(d) |


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| Pair | Key difference | Asked in |
|---|---|---|
| **Cable faults** | | |
| Murray vs Varley loop test | ratio arms only, balance by $P/Q$ vs extra variable resistor $S$ in the faulty leg | 20-21 5(b) |
| **HVDC and FACTS** | | |
| SVC vs STATCOM | thyristor-switched reactors and capacitors vs voltage-source converter, works at low voltage | 20-21 3(b) |
| Monopolar vs bipolar vs homopolar HVDC link | one conductor, earth return vs two conductors of opposite polarity vs two conductors of the same polarity, earth return | 21-22 2(a) |
| HVDC vs HVAC transmission | costly terminals, cheap lines, no stability limit vs cheap terminals, reactive power, stability limit | 21-22 6(b) |

# 10. Repeated questions:

8 questions came back, exactly or with new numbers; 1 of them in three or more papers. Most repeated first; the latest paper first in each row.

| No. | Times | Topic | Subtopic | Asked in (marks) |
|---|---|---|---|---|
| 1 | 3 | Line capacitance | Unsymmetrical spacing | 21-22 5(d) (10), 21-22 TT1 7(b), 20-21 1(c) (10) |
| 2 | 2 | Line inductance | Mutual inductance | 21-22 TT3 9(b) (5), 20-21 1(d) (10) |
| 3 | 2 | Line inductance | Composite conductors | 21-22 4(c) (8), 20-21 2(c) (10) |
| 4 | 2 | Corona | Critical disruptive voltage | 21-22 1(a) (2), 20-21 4(a) (5) |
| 5 | 2 | Power system stability | Equal area criterion | 21-22 6(c) (8), 20-21 6(b) (10) |
| 6 | 2 | Power system stability | Transient EMF | 21-22 6(d) (10), 20-21 6(c) (10) |
| 7 | 2 | Cable faults | Loop tests | 21-22 1(b) (5), 21-22 TT3 9(a) (5) |
| 8 | 2 | Underground cables | Dielectric stress | 21-22 2(c) (8), 21-22 TT2 8(b) (5) |

A sub-part with no marks after it had none printed on its paper.

# 11. Question style:

How each topic was asked, as a number of questions.

| Topic | Theory | Derivation | Numerical |
|---|---|---|---|
| Line inductance | 3 | 3 | 5 |
| Line capacitance | 1 | 3 | 1 |
| Corona | 3 | · | · |
| Overhead insulators | 3 | 1 | 3 |
| Mechanical design | 2 | · | 1 |
| Underground cables | 1 | 2 | 8 |
| Cable faults | · | · | 3 |
| Power system stability | 3 | · | 2 |
| HVDC and FACTS | 3 | · | · |

# 12. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

| No. | Topic | Marks | Covered so far |
|---|---|---|---|
| 1 | Underground cables | 85 | 26 % |
| 2 | Line inductance | 66 | 46 % |
| 3 | Overhead insulators | 48 | 61 % |
| 4 | Power system stability | 43 | 74 % |
| 5 | Line capacitance | 27 | 83 % |
| 6 | Cable faults | 20 | 89 % |
| 7 | Mechanical design | 17 | 94 % |
| 8 | HVDC and FACTS | 11 | 98 % |
| 9 | Corona | 8 | 100 % |


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